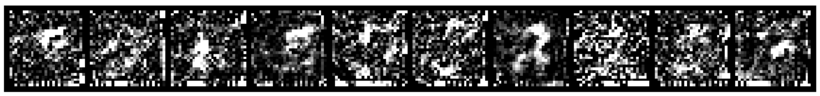
1 MNIST results

Following the reviewer's suggestion, we conducted additional experiments on MNIST [2]. We consider two training settings with 120 and 1200 samples, representing low-data and moderate-data regimes.

(a) RCLDM

(b) LDM



(c) DM

Figure 1: Comparison of results under different methods on 120 MNIST Images

As shown in Figure 1, when only 120 training samples are available, all methods are affected by the limited data. DM fails to produce meaningful digits, and LDM generates blurry and unstable results. In contrast, RCLDM is still able to produce recognizable digit structures.

(a) RCLDM

(b) LDM

(c) DM

Figure 2: Comparison of results under different methods on 1200 MNIST Images

When increasing the training set to 1200 samples (Figure 2), all methods improve, but the difference remains clear. RCLDM generates sharper and more consistent digits, while LDM still shows noticeable artifacts and DM remains unstable.

It is worth noting that in the 1200-sample setting, RCLDM only uses 120 anchor points. Despite using significantly fewer reference samples, it still achieves better visual quality, suggesting that the model can effectively leverage a small set of representative anchors.

2 New Baseline (MPGD)

We first tested MPGD in pixel space under the angle-conditioned generation setting. Here, the results were poor: MPGD without projection gave an FID of 329, and adding projection only changed this slightly to 327. In both cases, the generated images were essentially noise, with no recognisable object structure or meaningful control of pose. In other words, for the COIL lucky cat dataset in this limited data regime, pixel space MPGD did not produce usable samples, and the small difference in FID is not reflected in visual quality.

We then considered a latent-space adaptation of MPGD under the same conditioning setup. Starting from the vanilla latent diffusion baseline (FID 156), adding MPGD style guidance without projection gave a small improvement to 153, and including projection reduced the FID further to 147. This suggests that conditional guidance, together with latent space projection, can help in this setting. That said, the gain is still modest, and both latent space MPGD style variants remain clearly behind our Riemannian Corrected Latent Diffusion Model, which achieves an FID of 130. This is also consistent with the visual results: while the MPGD based samples broadly follow the target pose, they are noticeably blurrier and contain more local artefacts, whereas our method produces sharper and more faithful reconstructions.

Method	**FID ↓**
DM	352
LDM	156
MPGD$_p$	330
MPGD$_l$	153
RCLDM	**130**

Table 1: FID comparison on the Coil dataset (lower is better).

[1]

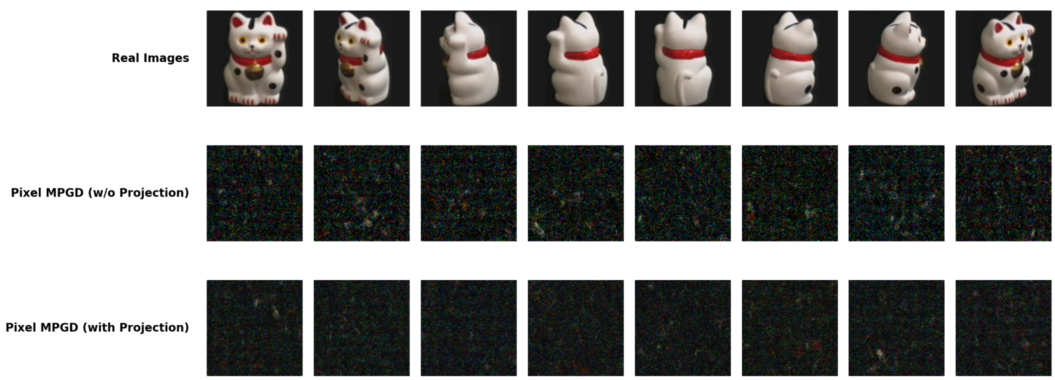


Figure 3: Coil Lucky Cat Pixel Space Reconstruction

Figure 4: Coil Lucky Cat Latent Space Reconstruction (AutoEncoder)

Table 2: FID comparison on the Coil dataset (lower is better).

Method	**FID ↓**
Pixel Space	
LDM	352
MPGD_p (w/o projection)	329
MPGD_p (with projection)	327
Latent Space	
LDM	156
MPGD_l (w/o projection)	153
MPGD_l (with projection)	147
RCLDM	**130**

3 Guidance Level

Following the reviewer's suggestion, we performed an additional ablation on the conditional COIL experiment by varying the classifier-free guidance scale over a range of strength values (e.g. $w \in \{1, 4, 6, 8\}$ for both conditional LDM and conditional RCLDM. We evaluate both conditional fidelity and sample quality under front/back-view conditioning.

The results show that increasing w strengthens conditional control for both methods, while RCLDM remains stable and consistently achieves lower FID than LDM across all guidance levels.

Table 3: FID comparison on COIL image dataset (condition: front)

Method	Guidance Level (w)	FID ↓
LDM	1	171.86
LDM	4	173.56
LDM	6	183.50
LDM	8	186.82
RCLDM	1	146.21
RCLDM	4	149.08
RCLDM	6	146.41
RCLDM	8	146.21

This provides empirical evidence that geometric correction does not conflict with strong guidance. Instead, it supports our interpretation that classifier-free guidance supplies a global task-driven direction, while the geometric correction acts as a local manifold regularizer.

4 Overfitting test

We conducted additional experiments on the COIL dataset to explicitly examine overfitting behaviour in pixel-space diffusion models.

We compute: (i) the minimum LPIPS between generated samples and training samples, and (ii) the minimum LPIPS between pairs of (nearest)training samples (as a reference for dataset similarity). For the diffusion model, we observe that the minimum LPIPS between generated and training samples is 0.03, which is lower than the minimum LPIPS between (nearest)training samples (0.034). This indicates clear signs of overfitting. However, despite this overfitting behaviour, the model still produces poor unconditional samples, with a high FID of 206.

In contrast, for RCLDM, the minimum LPIPS between generated samples and training samples is 0.34, which is significantly higher than the minimum LPIPS between training samples, indicating that the model does not simply memorize the dataset. At the same time, RCLDM achieves a much lower FID of 130, demonstrating better generative performance.

These results show that, although diffusion models can overfit in small-data regimes, such memorization does not translate into a stable or high-quality generative model. In contrast, our method benefits from the learned low-dimensional geometry and yields more stable behaviour in this setting. We will revise the manuscript to clarify this distinction and avoid the overly strong claim.

Method	FID ↓	Min LPIPS (Gen vs Train)	Min LPIPS (Train vs Train)
DM	206	0.03	0.034
RCLDM	**130**	0.34	0.034

Table 4: Overfitting analysis on COIL dataset.

References

[1] Yutong He, Naoki Murata, Chieh-Hsin Lai, Yuhta Takida, Toshimitsu Uesaka, Dongjun Kim, Wei-Hsiang Liao, Yuki Mitsufuji, J Zico Kolter, Ruslan Salakhutdinov, and Stefano Ermon. Manifold preserving guided diffusion. In *The Twelfth International Conference on Learning Representations*, 2024.

[2] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.